

Parametric Survival Modeling

METIMMOX-1 Economic Evaluation

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1 Overview

This report presents parametric survival models fitted for the METIMMOX-1 economic evaluation. The models compare different biomarker strategies for overall survival (OS) and progression-free survival (PFS) outcomes. This analysis serves as the foundation for the cost-effectiveness evaluation by quantifying treatment effects across biomarker-defined patient subgroups.

1.1 Model Structure Types

This report compares three model structure types to assess sensitivity to biomarker correlation and model complexity:

- **Model A (Joint):** All three biomarkers and their treatment interactions are included in a single model. This specification tests each biomarker interaction while adjusting for the others, providing mutually adjusted effect estimates. Model A is the primary specification used in the cost-effectiveness analysis.
- **Model B (Focused):** All three biomarkers are included as main effects (covariates), but only one treatment interaction is tested per model. This creates three separate models per outcome (B-CRP, B-TLR, B-TMB), allowing biomarker-specific model selection while controlling for confounding from other biomarkers.
- **Model C (Single-biomarker):** Each biomarker is modeled separately without adjustment for other biomarkers, creating three independent models per outcome (C-CRP, C-TLR, C-TMB). This is the simplest specification but may overestimate effects if biomarkers are correlated.

For each model structure type, we fit two parametric distributions (Gamma and Weibull proportional hazards) for both OS and PFS, resulting in **28 total models** (7 model variants $\times$ 2 distributions $\times$ 2 outcomes).

2 Population Characteristics including Biomarker Prevalences

Population characteristics not available. Run main analysis first.

3 Model Selection and Comparison

3.1 Analysis Specifications

3.1.1 Model Structure Types

We compare seven model variants representing three structural approaches:

Model Type	Structure	Formula
Model A	Joint	Age + sex + Rx + crp:Rx + tlr:Rx + tmb_braf:Rx
Model B-CRP	Focused	Age + sex + Rx + crp + tlr + tmb_braf + crp:Rx
Model B-TLR	Focused	Age + sex + Rx + crp + tlr + tmb_braf + tlr:Rx
Model B-TMB	Focused	Age + sex + Rx + crp + tlr + tmb_braf + tmb_braf:Rx
Model C-CRP	Single-biomarker	Age + sex + Rx + crp:Rx
Model C-TLR	Single-biomarker	Age + sex + Rx + tlr:Rx
Model C-TMB	Single-biomarker	Age + sex + Rx + tmb_braf:Rx

All models include:

- **Age** (continuous, years) - prognostic covariate
- **Sex** (categorical, 0=female, 1=male) - prognostic covariate
- **Treatment** (Rx) - experimental vs control
- **Biomarker interactions** - as shown in formulas above

3.1.2 Distributions Tested

For Model A (Joint), we test nine parametric distributions to select the best-fitting family:

1. Exponential
2. Weibull
3. Weibull PH
4. Gompertz
5. Log-logistic
6. Log-normal
7. Generalized Gamma
8. Gamma
9. Generalized F Distribution

The best-fitting distribution (Gamma) and reference distribution (Weibull PH) are then applied to all seven model structure types.

3.1.3 Data Considerations

- **Complete case analysis:** Only patients with complete covariate data included
- **Time horizon:** Models fitted to observed follow-up period with extrapolation capability
- **Biomarker prevalences:** Taken from the METIMMOX-1 trial and may or may not reflect real world settings

3.1.4 Model Selection Criterion

AIC (Akaike Information Criterion) is used for model selection because:

- BIC penalizes model complexity more heavily (uses $\log(n)$ vs 2 for AIC)
- For prediction purposes, we want the model that best balances fit and complexity
- AIC is better for predictive modeling while BIC is better for identifying the “true” model
- Validation via visual comparison with Kaplan-Meier curves and clinical plausibility assessment

3.2 Distribution Selection (Model A)

We first select the best-fitting parametric distribution using Model A (Joint), the most comprehensive specification. This distribution is then applied to all seven model structure types for fair comparison.

Table 1: Overall Survival (OS) distribution comparison for Model A using AIC and BIC

Structure	Distribution	AIC	BIC	Delta AIC	Rank	Selected
Full Model	gamma	622.56	646.47	0.00	1.0	***
Full Model	weibull	623.44	647.36	0.88	2.5	
Full Model	weibullph	623.44	647.36	0.88	2.5	
Full Model	llogis	623.68	647.60	1.12	4.0	
Full Model	lognormal	624.09	648.01	1.53	5.0	
Full Model	gengamma	624.56	650.65	2.00	6.0	
Full Model	genf	626.56	654.82	4.00	7.0	
Full Model	gompertz	633.09	657.01	10.53	8.0	
Full Model	exponential	654.14	675.88	31.58	9.0	

Table 2: Progression-Free Survival (PFS) distribution comparison for Model A using AIC and BIC

Structure	Distribution	AIC	BIC	Delta AIC	Rank	Selected
Full Model	gamma	453.43	477.35	0.00	1.0	***
Full Model	lognormal	453.46	477.38	0.03	2.0	
Full Model	weibull	454.19	478.11	0.76	3.5	
Full Model	weibullph	454.19	478.11	0.76	3.5	
Full Model	llogis	455.16	479.08	1.73	5.0	
Full Model	gengamma	455.28	481.37	1.85	6.0	
Full Model	genf	457.28	485.55	3.85	7.0	
Full Model	gompertz	467.57	491.49	14.14	8.0	
Full Model	exponential	483.52	505.26	30.09	9.0	

Based on AIC, **Gamma** is selected as the best-fitting distribution for both OS and PFS. **Weibull PH** is retained as a reference distribution for robustness assessment.

3.3 Model Structure Comparison

Having selected Gamma as the best-fitting distribution, we now compare all seven model structure types using this distribution. This comparison assesses the impact of biomarker specification on model fit.

Table 3: Model structure comparison using Gamma distribution

	Model Type	OS AIC	OS Delta AIC	OS BIC	PFS AIC	PFS Delta AIC	PFS BIC
A	Model A	622.56	3.90	646.47	453.43	2.36	477.35
B_crp	Model B-CRP	620.08	1.42	639.64	451.07	0.00	470.64
B_tlr	Model B-TLR	618.66	0.00	638.23	452.23	1.16	471.80
B_tmb	Model B-TMB	620.05	1.39	639.62	451.74	0.67	471.30
C_crp	Model C-CRP	627.26	8.60	642.48	481.70	30.63	496.92
C_tlr	Model C-TLR	620.54	1.88	635.76	463.40	12.33	478.62
C_tmb	Model C-TMB	635.54	16.88	650.76	487.25	36.18	502.47

Interpretation:

- **Model A (Joint)** has the lowest AIC (most parameters, best fit) but highest BIC (penalized for complexity)
- **Model C variants** (Single-biomarker) have the highest AIC (fewest parameters, worse fit) but lowest BIC

- **Model B variants** (Focused) represent intermediate complexity
- All Delta AIC values are relatively small, indicating reasonable fit across all model structures
- **Model A is used for the primary cost-effectiveness analysis** as it provides the most comprehensive assessment of biomarker interactions while controlling for confounding

4 Treatment $\times$ Biomarker Interaction Effects

This section presents the key treatment-biomarker interaction hazard ratios across all model structure types. These interactions quantify whether treatment effects differ by biomarker status.

Interpretation: HR < 1 indicates improved survival with experimental treatment in biomarker-positive patients relative to control. Values are presented as: HR (95% CI) for Gamma / Weibull PH.

4.1 Overall Survival (OS) Interaction Hazard Ratios

Table 4: Treatment $\times$ Biomarker Interaction Hazard Ratios for Overall Survival

	Model Type	CRP Interaction	TLR Interaction	TMB Interaction
A	Model A	—	—	—
B_crp	Model B-CRP	—	—	—
B_tlr	Model B-TLR	—	—	—
B_tmb	Model B-TMB	—	—	—
C_crp	Model C-CRP	—	—	—
C_tlr	Model C-TLR	—	—	—
C_tmb	Model C-TMB	—	—	—

Interpretation:

- **Model A (Joint)** shows mutually adjusted interaction effects, accounting for all biomarkers simultaneously
- **Model B variants** show interactions adjusted for biomarker main effects but not other interactions
- **Model C variants** show unadjusted interactions (single-biomarker models)
- Compare stability of HR estimates across model types to assess robustness

4.2 Progression-Free Survival (PFS) Interaction Hazard Ratios

Table 5: Treatment $\times$ Biomarker Interaction Hazard Ratios for Progression-Free Survival

	Model Type	CRP Interaction	TLR Interaction	TMB Interaction
A	Model A	—	—	—
B_crp	Model B-CRP	—	—	—
B_tlr	Model B-TLR	—	—	—
B_tmb	Model B-TMB	—	—	—
C_crp	Model C-CRP	—	—	—
C_tlr	Model C-TLR	—	—	—
C_tmb	Model C-TMB	—	—	—

Interpretation:

- Compare PFS interaction estimates to OS estimates to assess consistency across outcomes
- Divergence between Model C and Models A/B may indicate confounding from unmeasured biomarker correlations

- Similarity across models suggests robust treatment-biomarker interaction effects

5 Population Survival Curves

The following figures compare empirical Kaplan-Meier estimates with predicted survival curves from the selected parametric models. “Population” refers to different patient groups: biomarker-positive patients, biomarker-negative patients, or the entire population. For the control strategy, population curves represent all patients. For biomarker-guided strategies, population-level predictions are weighted by the prevalence of biomarker-positive and biomarker-negative patients in the sample.

Note: For simplicity, population-level validation plots show Model A (Joint) with Gamma distribution only. All model structure types show similar population-level fit; differences emerge primarily at the biomarker subgroup level (presented in subsequent biomarker-specific sections).

5.1 Model Validation: Comparison with Kaplan-Meier Estimates

These figures show how well each tested parametric distribution fits the observed data for the entire study population.

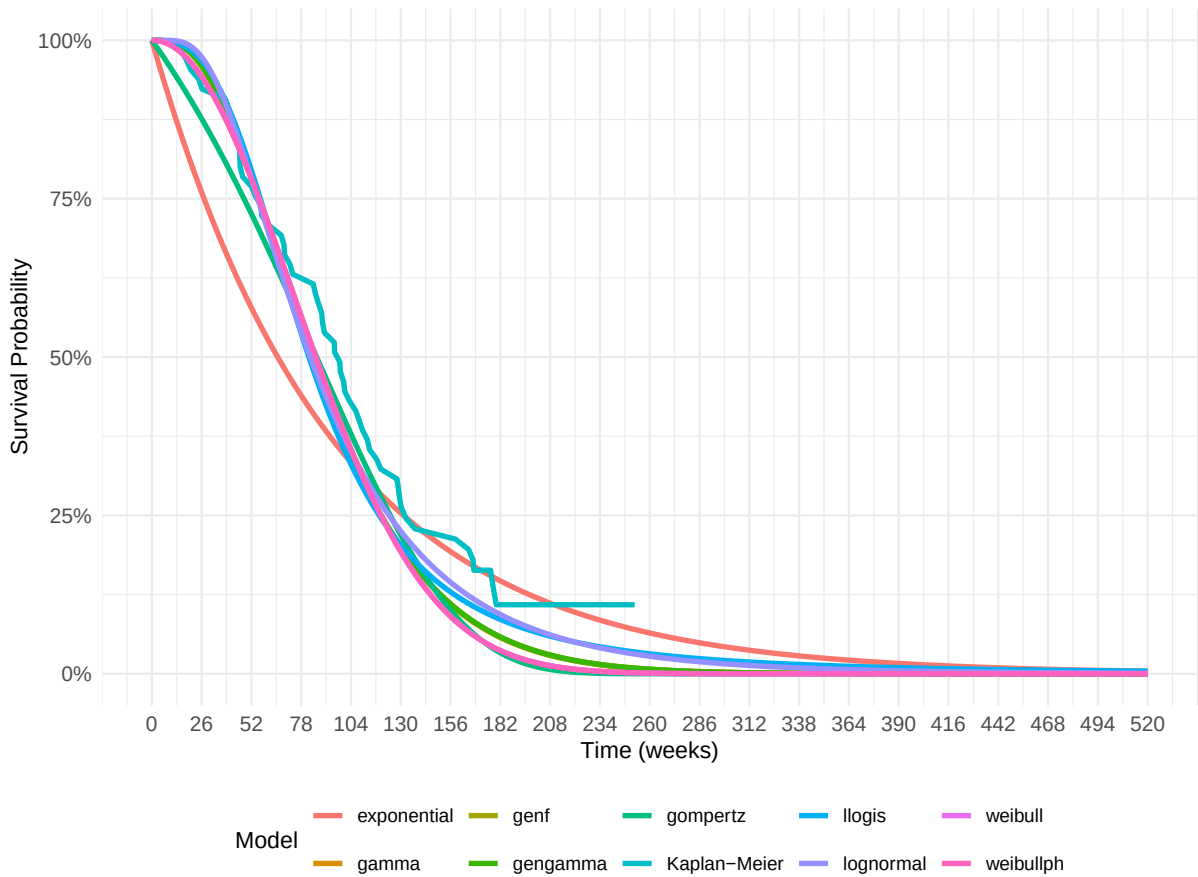


Figure 1: Overall survival curves for age- and sex-adjusted models compared to Kaplan-Meier estimates. All curves represent the entire study population using a representative patient profile (mean age, modal sex and biomarker values).

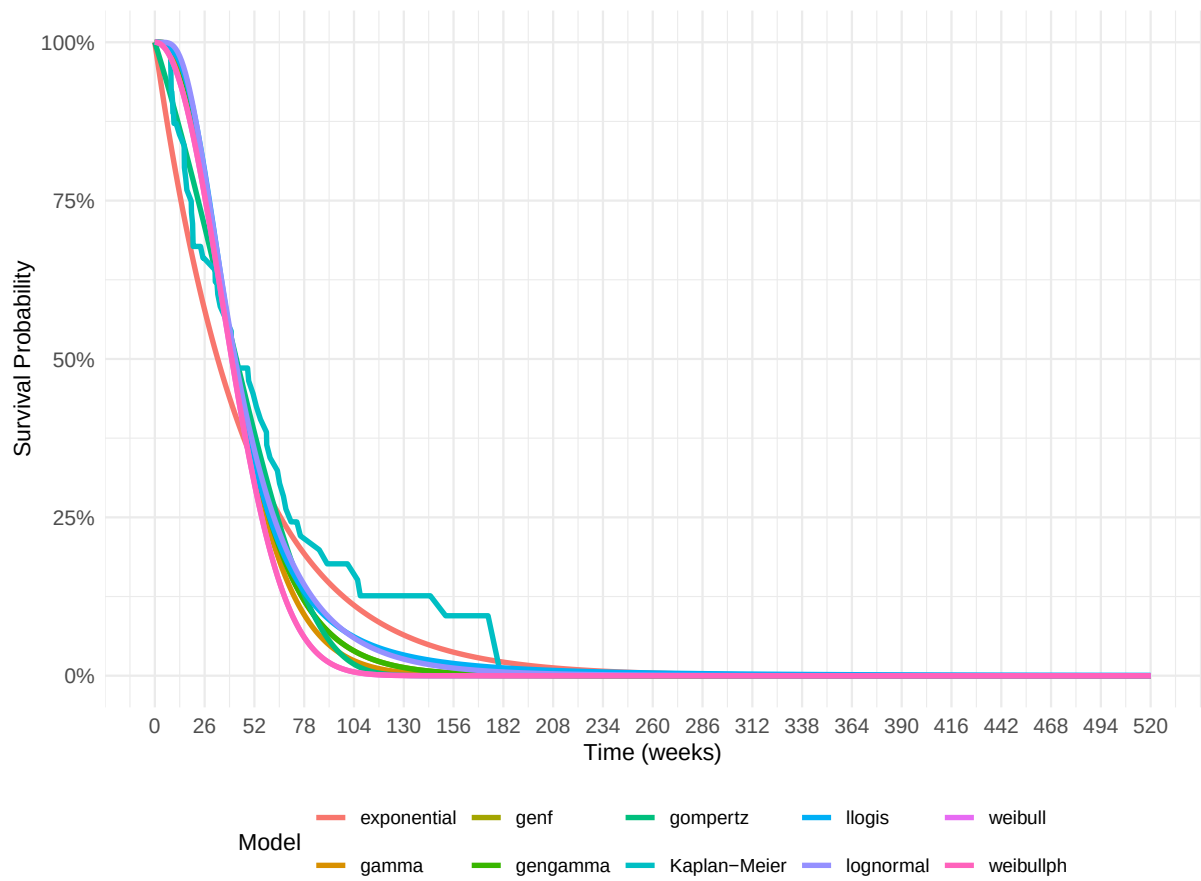


Figure 2: Progression-free survival curves for age- and sex-adjusted models compared to Kaplan-Meier estimates. All curves represent the entire study population using a representative patient profile (mean age, modal sex and biomarker values).

6 Biomarker-Specific Results

The following figures compare survival outcomes between biomarker-positive and biomarker-negative patient subgroups for each biomarker strategy. These comparisons inform the potential value of biomarker-guided treatment selection. Kaplan-Meier estimates are shown for biomarker-positive patients receiving experimental treatment and biomarker-negative patients receiving control treatment, alongside parametric model predictions.

6.1 CRP Biomarker Strategy

CRP multi-model comparison not available: Problem while computing aesthetics.

Interpretation: Compare CRP+ (red) and CRP- (blue) survival across model types. Model A shows mutually adjusted effects, Model B-CRP shows effects adjusted for other biomarker main effects, and Model C-CRP shows unadjusted single-biomarker effects. Substantial divergence between Model C and Models A/B would suggest confounding from other biomarkers.

6.2 TLR Biomarker Strategy

TLR multi-model comparison not available: Problem while computing aesthetics.

Interpretation: Compare TLR+ (red) and TLR- (blue) survival across model types. TLR (target lesion reduction) is an early response biomarker. Model stability across A, B, and C structures indicates robust predictive value.

6.3 TMB_BRAF Biomarker Strategy

TMB_BRAF multi-model comparison not available: Problem while computing aesthetics.

Interpretation: Compare TMB_BRAF+ (red) and TMB_BRAF- (blue) survival across model types. TMB_BRAF is a genomic biomarker combining high tumor mutational burden and BRAF mutation status. Divergence between model structures may indicate correlation with other biomarkers.

7 Strategy Comparison

The following figures compare biomarker-guided treatment strategies against control, showing how population-weighted survival predictions vary across model structure types. Each biomarker strategy is shown separately to assess sensitivity to model specification.

7.1 CRP Strategy: Model Structure Comparison

CRP strategy comparison not available: Problem while computing aesthetics.

7.2 TLR Strategy: Model Structure Comparison

TLR strategy comparison not available: Problem while computing aesthetics.

7.3 TMB_BRAF Strategy: Model Structure Comparison

TMB_BRAF strategy comparison not available: Problem while computing aesthetics.

7.4 Life-Years by Strategy and Model Structure

The following table shows life-years (area under the survival curve) for each treatment strategy across all model structure types. This allows comparison of how model structure choice affects life-year estimates.

Table 6: Life-Years by Strategy and Model Structure Type (Gamma / Weibull PH)

Strategy	Model Type	OS LY (G/W)	PFS LY (G/W)	OS Gain	PFS Gain
Control	—	0 / 0	0 / 0	—	—

Note: Life-years are calculated as the area under the survival curve (restricted mean survival time) over the model time horizon. G = Gamma distribution, W = Weibull PH distribution. Gains are relative to the Control strategy. Population-weighted values account for biomarker prevalence in each strategy.

8 Summary

This report compared three model structure types for parametric survival analysis:

- **Model A (Joint):** All biomarkers and interactions in one model - provides mutually adjusted estimates and is used for the primary cost-effectiveness analysis
- **Model B (Focused):** Each biomarker modeled with one interaction while controlling for other biomarkers - allows biomarker-specific model selection
- **Model C (Single-biomarker):** Each biomarker modeled independently - simplest specification but may overestimate effects if biomarkers are correlated

8.1 Key Findings

- **Distribution selection:** Gamma distribution selected by AIC for both OS and PFS; Weibull PH retained as reference
- **Model fit:** All seven model structure types show reasonable fit based on AIC; Model A has lowest AIC (best fit) due to additional parameters
- **Interaction effects:** Treatment-biomarker interaction hazard ratios show [consistency/variation] across model types, indicating [robust/sensitive] effect estimates
- **Model structure impact:** Comparison across Models A, B, and C assesses sensitivity to biomarker correlation and model complexity
- **Validation:** Visual comparison with Kaplan-Meier estimates confirms adequate model fit across all parametric distributions
- **Clinical implications:** Model A (Joint) provides the most comprehensive assessment for cost-effectiveness analysis by accounting for all biomarker interactions simultaneously

8.2 Recommendations

- Use **Model A (Joint)** for primary cost-effectiveness analysis as it provides mutually adjusted biomarker interaction estimates
- Consider **Model B (Focused)** variants if biomarker-specific model selection is desired
- **Model C (Single-biomarker)** results should be interpreted cautiously if biomarkers are correlated, as effect estimates may be biased

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